WHICH CONDITIONING RECORDING DOES A VOICE CLONE FOLLOW? A BALANCED CROSS-MICROPHONE CROSSOVER

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ABSTRACT

Voice-clone attribution usually stops at the generator or speaker. We ask a finer question: which of two same-speaker recording events supplied the conditioning recording? In a pre-specified complete crossover, a fixed 54-speaker VCTK roster provides two events with paired microphone captures. Four publicly available cloning systems and four fixed generation texts, distinct from the candidate transcripts, produce 3,456 clones. Generation uses one microphone; attribution compares the two events through the other, so speaker identity, generated text, exact waveform and a shared microphone fingerprint cannot decide the label. With fixed speaker-verification readouts, speaker-averaged two-alternative attribution accuracy for mic1→mic2 is .896 [.869,.921] for ECAPA and .622 [.593,.652] for WavLM; reversing the microphones gives .907 [.881,.931] and .620 [.589,.653]. When both candidates are themselves clones from a different system and a different text, attribution remains .805 [.780,.830] and .805 [.780,.831] for ECAPA. Brackets are 95% bootstrap intervals that resample whole speakers within this fixed, previously screened roster; systems and texts are not resampled. Attribution is evaluated only between two same-speaker candidates known to include the source event. Prompt interventions show that neither timing nor the long-term spectrum carries the cue and that the pitch and energy contour carries a minor share; the results do not establish whether an arbitrary recording was used.

Index Terms— voice cloning, source attribution, audio provenance, speaker verification

1. INTRODUCTION

Zero-shot speech generation conditions on a short reference recording. When that recording was used without consent, identifying the generator or cloned speaker does not answer the narrower provenance question: *which recording was used?* Existing source tracing attributes a synthesis pipeline [1, 2], and clone attribution identifies an enrolled speaker [3, 4, 5]. Recording-level evidence would instead connect an output to one conditioning event among other recordings of the same voice.

That target is easy to confound. A clone and candidate share speaker identity; one candidate may be intrinsically closer to every clone; recording session, duration or microphone can reveal the answer; and generated content may overlap. Prior prompt-transfer work shows preservation of prosody and acoustic environment [6, 7, 8], while speaker embeddings themselves encode session and channel [9, 10]. High retrieval from a same-speaker pool therefore does not by itself show that the target follows the recording used for conditioning.

We turn the question into a balanced intervention. For each speaker, we clone from recording event A and event B, then test both signed comparisons. Prompt and candidate are paired captures from different microphones. Four systems and texts are fully crossed. ECAPA-TDNN [11] (SpeechBrain `spkrec-ecapa-voxceleb`) is fixed as the primary encoder, with a speaker-verification-fine-tuned WavLM [12] (`microsoft/wavlm-base-plus-sv`) as a second embedding model for corroboration. The experiment uses a 54-speaker complement inherited from ECAPA-informed development rosters. All speakers in that complement are retained, and metadata alone determines pair selection without clone outcomes.

We make three contributions. (1) The complete A/B crossover tests whether a clone follows its conditioning recording after a pickup-path change. (2) The evaluation, whose analysis plan and decision rule were fixed before outcomes were inspected, covers 3,456 clones, 54 speakers, four systems and two fixed representations. (3) The boundary is explicit: recording-event attribution is consistent in a two-candidate closed set, but the experiment identifies neither the physical carrier nor deployable recording-presence detection.

2. RELATED WORK

Attribution and provenance. Generator tracing, shared-generator verification and speaker attribution operate above the individual recording [1, 2, 3, 4]. Recent source-verification and passive-fingerprint studies distinguish generating systems and examine attribution cues [13, 14, 15]; our label instead identifies the conditioning event within one speaker. Watermarking requires instrumentation before misuse [16]; VoiceMark attributes cloned speech through a watermark embedded in its inference-time reference [17], whereas our candidates are unmodified recordings. Membership inference asks whether audio entered training [18], not whether it was the inference-time prompt. We found no prior controlled study that intervenes on the prompt recording within speaker while changing the candidate pickup path.

Prompt transfer and nuisance information. Training-free generator attribution uses frozen representations [19, 20], and recording-related nuisance in speaker embeddings motivates cross-recording sampling [21]. Prompt-conditioned systems transfer prosody and environment [6, 7, 22], although cloning also alters style systematically [23]; prompt cosine is a standard synthesis metric [24]. Exact prosody cloning transfers reference prosody [25]. Source-voiceprint tracing recovers identity from converted audio [26], while rank/local-disclosure work treats linkage without global thresholds [27, 28]. None intervenes on which of two same-speaker recording events conditions a clone. Concurrent work separates prompt environment from identity [29]. These works motivate recording-level

Primary: opposite-pickup attribution

$A_{m1} \to c_{A,m1} \to \{A_{m2}, B_{m2}\}$ and

$B_{m1} \to c_{B,m1} \to \{A_{m2}, B_{m2}\}$

Correct labels are reversed across arms; speaker, system and generated text are held fixed. The direction-reversal analysis swaps $m1$ and $m2$.

Fig. 1. Complete two-arm crossover. Generation never consumes the microphone used for the attribution candidates.

Table 1. What the crossover blocks, and the residual scientific boundary.

Shortcut	Design response	Residual
Speaker identity	A/B within speaker	none for the label
Candidate preference	both signed arms	arm interactions
Generated content	four texts shared A/B	text fixed, not sampled
Exact waveform/device	opposite microphone	event acoustics
Duration/text length	metadata-matched A/B	other local cues

leakage but do not identify the chosen event. We target *A versus B*, not generic clone–prompt resemblance.

3. CROSS-MICROPHONE CROSSOVER

3.1. Task and design

For speaker s, let recording events A and B each have paired VCTK mic1/mic2 captures [30]: a DPA 4035 omni and a Sennheiser MKH 800. That the two are genuinely distinct signals is the design's load-bearing assumption. We tested it rather than assuming it. Raw correlation cannot settle the question. A delayed copy can score near zero, while genuine pairs reach .92 median once delay-aligned. We therefore use the normalized least-squares residual $\min_{g,|\tau|\le 50\,\mathrm{ms}} \|y - g\,T_\tau x\|_2/\|y\|_2$ over a scalar gain g and a zero-padded delay T_τ, where x and y are the mic1 and mic2 captures; its .00001 duplicate bar is a numerical tolerance for these transforms, not a universal perceptual threshold. Across all 108 event captures, the residual is .38 median (range .21–.70). Exact, gain-scaled and delayed injections on every source, including both delay-window boundaries, are at most .000000026; no pair is byte-identical. A clone $c_{A,m1}$ is conditioned on event A's mic1 capture. The primary comparison asks whether a fixed encoder f gives

$$I[\cos(f(c_{A,m1}), f(A_{m2})) > \cos(f(c_{A,m1}), f(B_{m2}))]\,, \qquad (1)$$

with ties worth $1/2$. We also clone from B and reverse the correct label. Thus an encoder's unconditional preference for A cannot create a positive effect. The predeclared direction reversal swaps microphones (mic2 prompt, mic1 candidates) and cannot rescue a failed primary direction.

With $q(u,v) = \cos(f(u), f(v))$ and $d_X = q(c_X, A) - q(c_X, B)$ on opposite-microphone candidates, a cell contributes $\{I[d_A > 0] + I[d_B < 0]\}/2$. Clone-independent candidate bias therefore yields chance; ties receive $1/2$.

The threat model is deliberately narrow. The analyst has a clone and two same-speaker candidate recordings; exactly one is the other-microphone capture of the event that supplied the conditioning recording. The objective is source attribution, not deciding whether an arbitrary recording was used. Encoder parameters, score direction and the tie rule are fixed before evaluation. The analyst does not train on these speakers or systems.

3.2. Frozen roster and generation

VCTK v0.92 documents DPA 4035 and Sennheiser MKH 800 microphones in a fixed hemi-anechoic setup [30]. Its public description does not establish acquisition timing, so our claim requires paired captures, not simultaneity. The frozen manifest independently supports the pairing: all 108 selected event/microphone pairs have identical frame counts and sampling rates.

The 54-speaker roster is the exact complement of two earlier development subsets. The first applied an ECAPA pair-separation screen to 105 speakers (pair cosine below that speaker's 90th percentile over pairs of duration-eligible utterances) and retained the lexicographically first 30 passing speakers. The second retained all 24 remaining speakers with one eligible pair at ECAPA cosine distance $\le$ .20 and another at $\ge$ .38, pair-mean durations within .50 s. The roster is therefore disjoint from those two subsets, although its speakers appeared in an earlier exploratory screen. It is blind to this experiment's clone outcomes, but not embedding-naive or population-sampled. Within that fixed complement, all 54 speakers remain. The A/B-pair rule below never inspects an embedding, score or clone outcome.

For each speaker, selection enumerates 4–10 s paired utterances and excludes the conditioning utterance used for that speaker in the earlier exploratory screen (listed in the public selection manifest). It requires an A/B duration gap of at most .25 s and a relative seconds-per-UTF-8-byte gap $|r_A - r_B|/[(r_A + r_B)/2]$ of at most 5% on *both* microphones. A deterministic metadata-only rule then takes the lexicographic minimum of the largest tolerance-normalized gap, their sum, the largest rate gap and the largest duration gap; no embedding, score or clone outcome enters within-roster pair selection. Transcripts A and B differ. Generation instead uses four fixed texts that are disjoint from both candidates and fully crossed across arms. These four texts were also used in our earlier experiments; they are fixed experimental factors. A post-hoc ASR screen (Whisper-large-v3-turbo) gives median WER .04 against the requested text; no clone (0 of 3,456) was flagged for four or more consecutive recognized words shared with its conditioning transcript after excluding sequences present in the requested text. This diagnostic does not establish the absence of reference-speech repetition. The selected roster is substantially tighter than the eligibility limits. Its worst duration gap is .112 s, and its worst relative duration-per-byte gap is 2.57% across either microphone.

We evaluate F5-TTS (f5-tts 1.1.22) [31], XTTS-v2 (coqui-tts 0.25.3) [32], CosyVoice 2 [33] and Seed-VC [34]; exact checkpoint revisions are listed in the public artifact. Seed-VC converts one fixed LibriTTS-R [35] source utterance per generation text, shared across targets. The full design is 54 speakers $\times 2$ events $\times 2$ prompt microphones $\times 4$ texts $\times 4$ systems $=$ 3,456 clones. Exact checkpoints, repositories, environments, files and per-clone ancestry ledgers are hash-pinned.

3.3. Estimand and decision criteria

Each speaker contributes 32 dependent comparisons per direction (two arms, four systems, four texts). We first average within speaker, then give all 54 speakers equal weight. Percentile intervals resample whole speaker vectors 100,000 times with a frozen seed. They quantify stability to speaker composition in this fixed eligible roster, not population-coverage confidence. Systems and texts are fixed crossed factors; system-specific points are descriptive and receive no individual inferential claims.

The primary success criteria, fixed before outcomes were inspected, were ECAPA accuracy $\ge$ .80 with lower stability-interval

Table 2. Speaker-averaged two-alternative attribution accuracy [95% speaker-resampling stability interval]; 1,728 comparisons per direction (clone-to-clone: 15,552). Same-mic columns are diagnostics, never verdict inputs; clone-to-clone candidates come from a different system and text.

Prompt→candidate	Encoder	Cross-mic	Same-mic diagnostic	Clone-to-clone
mic1→mic2	ECAPA	**.896 [.869,.921]**	.938 [.920,.955]	.805 [.780,.830]
	WavLM	**.622 [.593,.652]**	.631 [.602,.660]	.610 [.595,.626]
mic2→mic1	ECAPA	**.907 [.881,.931]**	.930 [.910,.948]	.805 [.780,.831]
	WavLM	**.620 [.589,.653]**	.659 [.627,.692]	.612 [.593,.632]

endpoint $> .70$, and WavLM lower endpoint $> .50$; the .80/.70 rule is an author-declared large-effect criterion, not a deployment-utility claim. Both reversed-microphone estimates are reported; we claim corroboration after reversal only if both lower endpoints exceed .50. Diagnostics cannot rescue any criterion.

We fixed the roster, analysis settings and success criteria before inspecting outcomes. A second, independent recomputation rehashed 3,456 clones and ledgers plus 216 candidates and reproduced every result field to 2×10^{-15}. The released artifact contains per-clone cosine scores with candidate identities and an independent verifier that reconstructs the 3,456-row identity grid, bootstrap and verdict without audio or model weights. Its execution environment is pinned.

4. RESULTS

4.1. The conditioning event survives pickup change

Table 2 reports the predeclared directions. ECAPA attributes the conditioning event in 89.6% of primary comparisons and 90.7% in reverse. It therefore meets the .80/.70 primary bar. WavLM is much weaker but directionally consistent at 62.2%/62.0%. All five predeclared criteria pass: the four lower endpoints in Table 2 (.869, .593, .881, .589) exceed .50, the primary ECAPA point .896 reaches .80 and its lower endpoint .869 exceeds .70. The ECAPA effect meets the frozen large-effect rule, and WavLM corroborates its direction.

Clone candidates. A stricter comparison, whose grid and decision rule were fixed before analysis, replaces both real candidates by clones: the query clone is compared with two clones of the same speaker made by a different system from a different text, conditioned on events A and B through the opposite microphone (288 comparisons per speaker and direction). The correct candidate shares only the conditioning event. Attribution remains .805 [.780,.830] and .805 [.780,.831] for ECAPA and .610 [.595,.626] and .612 [.593,.632] for WavLM (Table 2, last column), so the event information survives a change of generator, output text, microphone and candidate medium.

Same-microphone diagnostics are only modestly higher: exact pickup identity is unnecessary, though paired captures of one event do not establish microphone invariance.

Figure 2 exposes the heterogeneity hidden by pooled points. All 54 primary ECAPA speaker means exceed .50; in reverse, 53 exceed it and one equals it. WavLM is not uniformly positive: 47 primary and 43 reverse speaker means exceed .50. Its role is therefore aggregate corroboration, not a per-speaker guarantee. Whole-speaker resampling preserves this heterogeneity rather than treating 1,728 comparisons per direction as independent.

4.2. Breadth across fixed systems

Table 3 gives descriptive system points. ECAPA exceeds .82 throughout, while WavLM spans .572–.671 and has weaker XTTS-

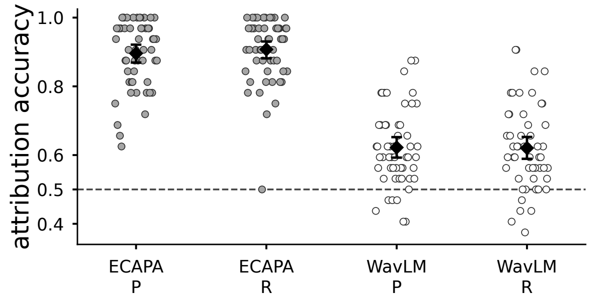

Fig. 2. Fixed-roster speaker means (54 dots/group; 32 comparisons per dot). Diamonds and bars are pooled points and 95% speaker-resampling stability intervals; the dashed line is chance. P/R denote primary/reverse directions.

Table 3. Descriptive attribution-accuracy points by system. P/R: primary/reverse.

Encoder, dir.	F5-TTS	XTTS-v2	CosyVoice 2	Seed-VC
ECAPA, P	.942	.833	.910	.898
ECAPA, R	.942	.822	.942	.924
WavLM, P	.618	.572	.671	.627
WavLM, R	.657	.574	.657	.593

v2 cells. The systems are fixed, and there are no simultaneous per-system intervals. The breadth claim is therefore limited to this tested design.

The ECAPA–WavLM gap rules out near-ceiling magnitude for every representation; WavLM's positive lower endpoints still oppose ECAPA-weight specificity. Both are speaker-verification embeddings, so a shared cue remains possible. Mean correct-minus-incorrect cosine margins are positive but small, ECAPA .079/.081 and WavLM .011/.013 for primary/reverse, and are descriptive only.

4.3. Descriptive sensitivity analyses

Table 4 stratifies the inherited ECAPA-informed complement by the earlier pair-separation screen. Direction persists among 36 pass-but-not-retained and 15 fail speakers. Three speakers absent from that record remain separate. This post-hoc check cannot create an embedding-naive sample or alter the pooled verdict.

Only the crossover and the clone-candidate comparison have decision rules fixed before analysis; the remaining analyses are post-hoc and descriptive. Two such extensions bound the claim. Enlarging the candidate set with metadata-matched captures of the same speaker through the opposite microphone, ECAPA rank-1 accuracy falls from .896/.907 at two candidates to .635 [.545,.720]/.587 [.510,.659] at sixteen (chance .0625; the 22 speakers with enough eligible captures). Re-synthesizing each primary-direction clone with a different system and a different text, the second-generation clone is still attributed to the original event at .796 [.759,.832] (ECAPA) and .596 [.568,.626] (WavLM).

Arm-separated points also oppose a one-candidate shortcut. For A/B, ECAPA is .889/.903 in primary and .892/.922 in reverse; WavLM is .588/.656 and .660/.581, respectively. Thus WavLM's arm asymmetry reverses with microphone direction. Descriptively, all 32 system×direction×encoder×arm cells are above .50 (minimum .523). No cell has a separate interval or multiplicity-adjusted test.

Table 4. Post-hoc points by status in the earlier pair-separation screen. P/R: microphone direction; no stratum-specific confirmatory claim.

Screen status	n	ECAPA P/R	WavLM P/R
Pass, not retained	36	.908/.907	.609/.596
Fail	15	.879/.915	.654/.685
Not evaluated	3	.833/.875	.625/.583

Table 5. Within-speaker one-to-one verification with one global threshold: EER and normalized minimum DCF. Target: clone versus its own event; non-target: clone versus the same speaker's other event; 1,728 each per direction. $C_{miss} = C_{fa} = 1$.

Encoder	Prompt→candidate	EER	minDCF, prior .01	minDCF, prior .5
ECAPA	mic1→mic2	.337	.980	.667
ECAPA	mic2→mic1	.343	.981	.678
WavLM	mic1→mic2	.447	1.000	.884
WavLM	mic2→mic1	.451	.998	.889

4.4. What changed relative to earlier evidence

An earlier exploratory two-reference crossover (30 speakers, 960 trials) obtained .806 [.757,.854] on one pickup path with favorable reference pairs and missed its frozen .90 point bar. We retain that as positive discovery, not confirmation.

The present experiment is a separate test: the 54-speaker development complement, all speakers retained, within-roster metadata-only A/B pairs, balanced labels, opposite-microphone candidates and fixed readouts. Its .80/.70 ECAPA bar was frozen for this design; passing it does not retroactively change the earlier .90 miss.

5. BOUNDARY AND LIMITATIONS

Attribution is not presence detection. The crossover is a closed-set two-alternative question known to contain the source. Table 5 sweeps one common threshold per encoder over the same scores, pooled over systems and arms, as one-to-one verification within speaker. At the .01 prior, minimum DCF sits at or near the reject-all cost of 1.000; whole-speaker resampling as in Table 2 gives ECAPA upper endpoints .989/.989. At the .5 prior of the two-candidate threat model it is .67/.68 and .88/.89. This post-hoc check is descriptive. A score can order A above B while no global threshold separates the two.

We therefore restrict investigative triage to a known-positive two-recording set for human provenance review, where the score can prioritize candidates. It cannot decide whether an arbitrary queried recording was present, and it is not an automated accusation.

The carrier is bounded, not identified. Opposite microphones exclude an exact waveform and shared device fingerprint, while A/B balancing excludes a fixed candidate preference, and metadata matching sharply limits duration shortcuts. Three post-hoc prompt interventions, synthesized with the same systems and scored against unmodified candidates (primary direction), bound what remains: equalizing the A/B durations or their long-term spectra leaves ECAPA attribution unchanged (.896 and .893 against .896), whereas flattening the pitch and energy contours lowers it to .840 [.804,.874] (.866 when the candidates receive the same transform). Static spectral coloration and timing therefore carry no measurable share, the prosodic contour a minor one, and at least .84 survives all three, bundled in fine spectral detail of the realization, articulation and their interaction with model conditioning. A positive mechanism claim requires interventions on those factors.

Scope. The result concerns four fixed checkpoints, four fixed texts, read English speech and one inherited 54-speaker VCTK complement. The complement is disjoint from the earlier development rosters but ECAPA-informed through their composition, not globally unseen or population-sampled. Checkpoint training corpora are not documented at recording level, so VCTK exposure cannot be excluded. The intervals do not sample systems, texts or a speaker population. Spontaneous conversational speech, noisy field recordings, other pickup paths, more than two candidates and open-set search remain untested.

Release and misuse. Code, score-level data, provenance metadata and aggregate/per-speaker results are at `https://github.com/rvirgilli/voice-clone-cross-microphone-crossover/tree/f2-icassp2027-rc2`; model weights, source audio and playable impersonations are not redistributed. Recording attribution can assist provenance audits, but it can also deanonymize speakers. Licensing alone does not settle consent.

6. CONCLUSION

In this fixed roster, clones preferentially match the other-microphone capture of their conditioning event across the two tested microphone paths. ECAPA meets the frozen large-effect criterion, and WavLM shows a smaller effect in the same direction. That direction persists when the microphone roles are reversed, when both candidates are clones from another system and text (.805), and after a second synthesis (.796). The primary and reverse pooled estimates weight the four fixed systems equally. This is a large closed-set signal between two tested paths that neither timing nor long-term spectral coloration carries; it does not identify the carrier positively and does not solve open-set recording-presence detection.

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